Project Design Phase-I
Solution Architecture

Date	08 March 2026
Team ID	699d5f9ac40b5ad0e9ee5a30
Project Name	QUOTES RECOMMENDATION CHATBOT USING NLP
Maximum Marks	4 Marks

Solution Architecture:

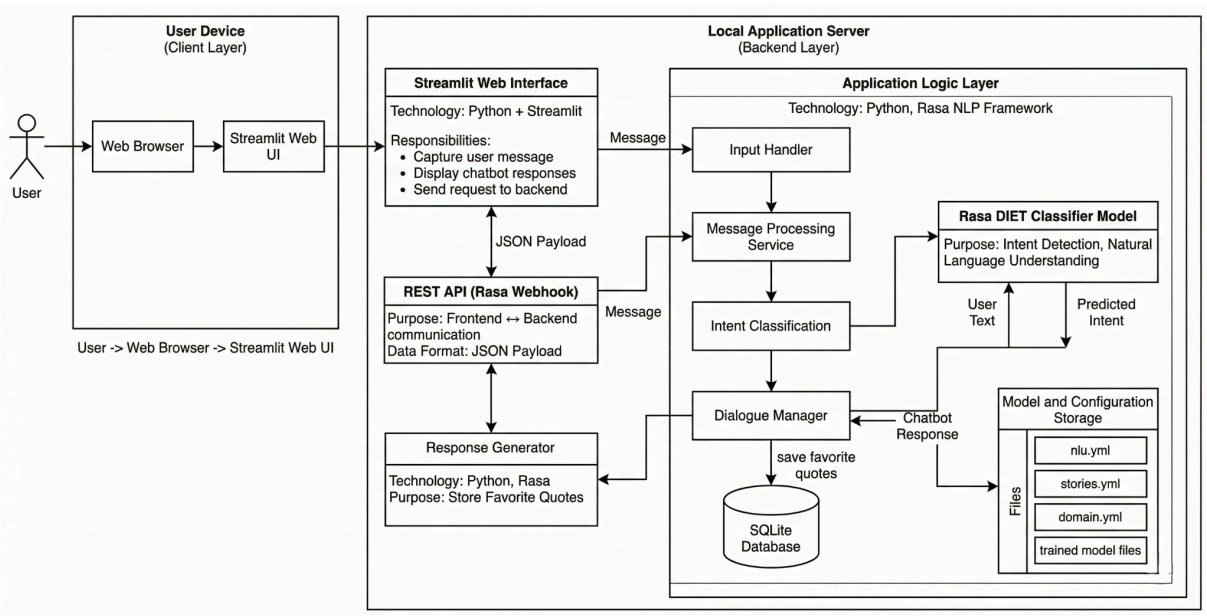

Figure 1: Architecture and data flow of the voice patient diary sample application

The solution architecture of the Quotes Recommendation Chatbot consists of a web-based user interface developed using Streamlit. Users interact with the chatbot by sending text messages through the web interface. These messages are transmitted to the backend NLP engine using a REST API.

The backend is implemented using the Rasa framework, which processes the user input using Natural Language Processing techniques. The Rasa DIET classifier model analyzes the text and identifies the user's intent.

Based on the predicted intent, the dialogue manager selects an appropriate quote response from predefined responses stored in the domain configuration. The system also supports saving favorite quotes using an SQLite database.

Finally, the generated response is sent back to the frontend interface and displayed to the user through the chat interface. This architecture enables an interactive and scalable conversational system for providing motivational and inspirational quotes.